Transfer: gravity and motion

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply gravity, projectile and orbital reasoning to a new set of cases.

SUCCESS CRITERIA - I can...

1. State a model and assumptions before calculating.
2. Connect diagrams to forces and motion.
3. Check units and explain the limits of conclusions.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “model” and “gravity” to explain your model or result. Sentence starters are optional.

model: a drawing, explanation or equation for a physical situation

gravity: an attractive interaction between masses

projectile: an object moving after launch under the stated forces

01 Warm-up: bring back an earlier idea

A circular orbital radius becomes 9 times as large around the same source.
Compare orbital speed and period.

Word help: radius = distance from the center to the edge of a circle or sphere

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F_g = mg$; $a_c = v^2/r$; $h = \frac{1}{2}gt^2$; range = $v_x t$.
Match each equation to its case: near-surface weight, circular motion, or horizontal launch. Use $g = 10\ m/s^2$.

Worked reasoning: A toy horizontal projectile falls for 2 s while moving at 4 m/s. Range is 8 m. The vertical motion sets time; the horizontal motion sets range once time is known.

Your turn: Draw two coordinated timelines, one for each direction, with a shared 2 s endpoint.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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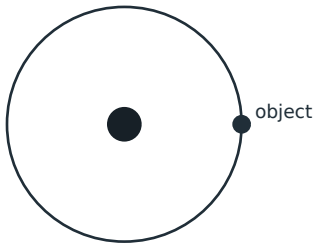
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A satellite moves in a circular orbit.
Draw velocity and gravity arrows.
Explain why floating does not mean zero gravity.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: 1 2 3

04 Apply the physics

A ball launches horizontally at 6 m/s from 20 m.
Find time, range, final vertical velocity and final speed.
Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A Sun-orbit returns every 2 years. Why can period alone not prove an Earth encounter?
Name one piece of evidence you would request.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

For fixed masses, what separation multiplier makes gravity 1/16 of its original value? What multiplier is needed if both masses also become 6 times as large?
Label the masses, field or force relevant to the question.

Word help: separation = distance between the specified objects or centers

Sentence starter: The ... in my model represents ... . Its label shows ... .



Label separation if needed.

Source Object

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

For GM=400 m³/s² and circular r=25 m, find speed, acceleration and period.
Explain what changes in velocity.
Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A horizontal projectile launches from a 45 m platform at 12 m/s; g=10 m/s².
Find flight time, range and final velocity. Then halve g with all initial conditions fixed and compare range.
Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Sentence starter: Both … . They differ in … because … .



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3